ment only approximates direct disalignment. Thus, it would not be unexpected for emulated disalignment to be less effective than direct disalignment. However, contrary to expectations, Figure 5 reveals that, in practice, emulated disalignment produces more harmful responses than direct disalignment, even when $\alpha = 1$. This is notable given the resource-intensive nature of direct disalignment (see Appendix B.1 for the training computations for direct disalignment). A qualitative comparison between emulated disaligned models' responses and direct disaligned models' responses to the same queries is in Appendix B.2.

However, these findings *do not* imply that emulated disalignment is *always better* than direct disalignment. Under large β^{-1} where the safety-aligned models are the safest, emulated disalignment underperforms direct disalignment by a great margin, and $\alpha > 1$ makes this performance degradation even more evident. We suspect this is because **optimizing for harmfulness sufficiently requires nuanced sequence-level adaptation for which a training-free token-level approximation is suboptimal**. Appendix B.2 illustrates the failure cases observed at $\beta^{-1} = 8$, where the emulated disaligned models tend to produce brief responses that limit their potential for harmfulness.

In summary, these synthetic experiments show that emulated disalignment can be competitive with resource-intensive direct disalignment, and making the models safer generally increases their risks of misuse under adversarial manipulation. However, when the safety-aligned models are sufficiently optimized for safety, ED generally require smaller α to be effective (e.g., 1/4 as shown in Figure 5). This agrees with our experiments on open-source models where we typically use $\alpha \ll 1$ for good empirical results (Section 5).

7 Conclusion

This study presents emulated disalignment (ED), an inference-time attack method that reverses safety alignment by contrasting the output token distribution of a safety-aligned language model against its pre-trained version, effectively producing a harmful language model without additional training. The finding that safety alignment might unintentionally promote harmfulness under simple adversarial manipulation should prompt the community to reconsider the open accessibility of language models even if they have been safety-aligned.

8 Limitations and Future Works

Our results also raise several questions that are out of the scope of this study: (1) Does ED transfer to black-box LLMs? Some black-blox LLMs (e.g., GPT-4) do allow limited visibility into their output token distributions, such as showing the log-likelihoods of the top-5 tokens. We wonder if this limited transparency is enough to undo efforts to make these models safe. (2) Can ED be used to attack model pairs with different vocabularies? While vanilla ED may not work directly, techniques like cross-vocabulary test-time search might help (Zhou et al., 2024). (3) How to defend against ED? For open-source models, this may involve designing more robust alignment algorithms during training. For closed-source models, the challenge is balancing output token distribution transparency with the risk of misuse. Additionally, ED is not just a risk; it can be a valuable data augmentation tool, generating synthetic harmful data to aid safety alignment. ED also has applications beyond attacking language models, including attacking safety-aligned visual language models, text-to-image diffusion models, and other generative models.

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Warning: The appendix contains samples that may be offensive or harmful.

A Experiments on Open-Source Language Models

A.1 Models

The table below lists links to all the language models used in this study, presented in pairs: each family consists of a pre-trained model followed by its safety-aligned version.

Models		HuggingFace Link
Llama-1	`Llama-1-7b`	https://huggingface.co/huggyllama/llama-7b
	`Vicuna-7b`	https://huggingface.co/lmsys/vicuna-7b-v1.3
Llama-2	`Llama-2-7b`	https://huggingface.co/meta-llama/Llama-2-7b-hf
	`Llama-2-7b-chat`	https://huggingface.co/meta-llama/Llama-2-7b-chat-hf
Mistral	`Mistral-7b`	https://huggingface.co/mistralai/Mistral-7B-v0.1
	`Mistral-7b-Instruct`	https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.1
Alpaca	`Alpaca-7b`	https://huggingface.co/PKU-Alignment/alpaca-7b-reproduced
	`Beaver-7b`	https://huggingface.co/PKU-Alignment/beaver-7b-v1.0

Table 3: Model links

Llama-1 family. `Llama-1-7b` (Touvron et al., 2023a) is a foundation model pre-trained on 1.4T tokens of public data. `Vicuna-7b` (Chiang et al., 2023) is a chat model fine-tuned from `Llama-1-7b` to emulate ChatGPT, focusing on helpfulness and safety.

Llama-2 family. `Llama-2-7b` (Touvron et al., 2023b) follows the `Llama-1-7b` training pipeline but with longer context length and group-queried attention. `Llama-2-7b-chat` (Touvron et al., 2023b) is an official chat model based on `Llama-2-7b`, optimized for helpfulness and safety.

Mistral family. `Mistral-7b` (Jiang et al., 2023) is a foundation model that claims to outperform `Llama-2-13b` on a broad range of benchmarks. `Mistral-7b-Instruct` (Jiang et al., 2023) is an official fine-tuned version of `Mistral-7b` that depends on system prompt to reinforce safety.

Alpaca family. `Alpaca-7b` (Taori et al., 2023) is fine-tuned from `Llama-1-7b` on 52K instruction-following demonstrations without safety guardrails. `Beaver-7b` (Ji et al., 2024) builds upon `Alpaca7b` and enhances safety through RLHF (Stiennon et al., 2020).

A.2 Methods and Baselines

Hyperparameters. During inference, we set the sampling temperature to 1 and limit the maximum number of generated tokens to 256 for all experiments. We use FlashAttention-2 (Dao, 2023) to accelerate language model inference. All model inferences are conducted on A100 GPUs. For ED, we use fixed α for each model family across different evaluation datasets: $\alpha = 0.6$ for the Llama-1 family, $\alpha = 0.3$ for the Llama-2 and Mistral families; and $\alpha = 2.0$ for the Alpaca family. The Alpaca family affords a greater α because the pre-trained model `Alpaca-7b` is an instruction-tuned model and thus more resilient to deviation. These hyperparameters were determined early in the project through trial and error using fewer than 50 harmful queries we devised while interacting with the emulated disaligned models. None of these hyperparameters are tuned on the evaluation datasets.

Zero-shot prompt template for pre-trained models. For foundation models pre-trained on the internet-scale corpus, we use a zero-shot prompt template (URIAL-style (Lin et al., 2023a)) so that the continuation of the text produces the responses to the user query:

```
# Instruction

Below is a conversation between a human and an AI assistant (you).
```